

ST AUGUSTINE BUOY 41012, USA

WMO: 996350

Lat: 30.042N

Lon: 80.534W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 02-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.3	47.8	28.7	22.7	49.8	31.0	25.2	50.5	35.8	58.4	33.5	56.7	22.2	330	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.5	85.3	78.7	84.5	78.2	84.1	78.0	81.0	82.7	80.1	82.8	79.3	82.7	10.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	158.2	82.0	79.2	151.4	81.9	78.2	146.2	82.1	44.4	82.8	43.4	82.7	42.7	82.8	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	31.8	26.9	24.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.7	62.3	62.6	66.0	70.1	74.9	79.4	81.0	81.6	80.7	76.7	70.3	66.2
	DBStd	8.23	7.22	5.87	5.49	3.65	2.33	1.71	1.61	2.08	1.72	4.08	4.96	6.26
	HDD50	14	8	3	1	0	0	0	0	0	0	0	0	2
	HDD65	383	133	100	56	9	0	0	0	0	0	3	18	64
	CDD50	8299	390	355	497	602	772	882	960	981	921	827	608	504
	CDD65	3195	50	32	88	162	307	432	495	516	471	365	176	101
	CDH74	24430	11	1	7	104	1146	3970	5231	5799	5026	2709	368	58
	CDH80	4120	0	0	0	0	2	357	1008	1577	999	177	0	0
Wind	WSAvg	13.3	15.5	14.8	13.8	13.0	12.6	11.1	10.4	11.0	12.8	13.6	15.7	15.3
Precipitation	PrecAvg	52.7	4.0	3.6	4.0	3.0	3.3	6.7	6.3	6.6	5.0	4.0	2.6	3.5
	PrecMax	66.7	11.5	10.4	10.0	6.6	9.1	10.0	11.2	12.6	13.1	11.9	5.6	8.3
	PrecMin	39.9	1.1	0.9	0.6	0.8	0.7	2.3	2.6	3.7	1.5	0.4	0.4	1.2
	PrecStd	7.5	2.2	2.0	2.2	1.6	2.0	2.1	1.9	1.9	2.4	2.7	1.5	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	73.3	75.2	77.1	80.3	84.5	85.5	86.5	84.9	83.5	79.5	76.6
		MCWB	69.7	68.8	71.1	68.0	74.1	78.0	79.0	79.5	77.3	77.2	73.1	71.2
	2%	DB	74.2	72.0	73.8	76.2	79.4	83.4	84.4	85.7	84.2	82.4	78.4	75.5
		MCWB	68.7	67.0	68.0	68.4	73.9	77.2	78.3	79.0	76.9	76.1	71.7	70.4
	5%	DB	72.9	70.8	73.0	75.3	78.6	82.5	84.0	85.1	83.7	81.8	77.3	74.7
		MCWB	67.4	65.5	66.9	68.5	73.1	76.8	78.1	78.8	76.6	75.5	70.3	69.3
	10%	DB	71.5	69.5	72.1	74.5	78.0	82.0	83.4	84.5	83.2	81.1	76.1	73.6
		MCWB	65.8	64.0	66.2	68.0	72.5	76.5	77.8	78.6	76.3	74.7	68.8	67.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.3	69.6	72.3	72.6	76.4	80.0	82.3	82.4	79.4	78.6	75.3	73.5
		MCDB	74.6	72.2	74.2	74.7	79.0	82.8	82.5	83.3	83.0	82.3	78.4	75.4
	2%	WB	70.2	68.3	69.6	71.5	75.4	79.1	80.7	81.4	78.7	77.7	73.5	72.2
		MCDB	73.3	71.0	72.3	74.1	78.4	81.7	81.9	83.1	82.7	81.5	77.7	74.6
	5%	WB	68.3	66.6	68.7	70.5	74.5	78.4	79.8	80.4	78.2	77.0	71.8	70.9
		MCDB	72.0	69.6	71.8	73.9	77.7	81.3	81.9	83.3	82.5	81.1	76.3	73.7
	10%	WB	66.9	64.7	67.7	69.5	73.6	77.7	79.0	79.6	77.6	76.1	70.3	69.5
		MCDB	71.1	68.3	71.2	73.6	76.9	80.8	81.8	83.4	82.2	80.6	75.5	72.8
Mean Daily Temperature Range	5% DB	MDBR	7.1	6.6	5.4	4.6	3.4	3.3	3.1	3.5	3.3	3.8	5.3	6.7
		MCDBR	5.3	5.7	4.5	4.2	3.7	3.5	3.2	3.2	3.4	3.6	5.3	5.4
		MCWBR	6.3	6.8	4.5	4.9	4.0	3.4	2.9	2.9	3.3	3.7	6.2	6.5
	5% WB	MCDBR	5.7	5.9	4.4	4.6	3.7	3.5	2.9	3.4	3.3	3.8	6.2	5.9
		MCWBR	6.5	7.4	5.0	5.7	4.0	3.8	3.4	3.6	3.2	4.0	7.5	7.2
Clear-Sky Solar Irradiance	taub	0.347	0.358	0.375	0.398	0.434	0.465	0.483	0.471	0.427	0.404	0.365	0.354	
	taud	2.539	2.516	2.473	2.390	2.332	2.314	2.269	2.307	2.419	2.453	2.508	2.540	
	Ebn at Noon	281	288	289	285	273	263	258	260	269	268	272	273	
	Edh at Noon	28	31	34	38	41	41	43	41	36	32	28	27	
All-Sky Solar Radiation	RadAvg	982	1209	1603	1990	2136	2085	2086	1919	1602	1378	1073	895	
	RadStd	74	122	126	99	135	185	107	115	136	108	72	73	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	+0.79	+1.37	N/A	N/A	N/A	N/A	-16	-112	+288	+209

Nomenclature: See separate page